

Re-scoped Proposal — DNA KEY, Nutritional Intervention, Homirn Ring ↔ Patent & Practitioner Portals

Version: Draft v0.1

Prepared for: CH Labs / Intrinsic

Prepared by: Motivity Labs (proposed)

Scope change: Excludes PIRTM, AProof, and OLHES. Focuses on integrating **DNA KEY**, **Nutritional Intervention**, and the **Homirn Ring** with the **Patent Portal**, and wiring the **Patent Portal** to the **Practitioner Portal**.

1) Objectives

1. Integrate the Homirn Ring data stream into the platform and surface actionable nutrition-centric insights to practitioners.
 2. Connect patient/practitioner records with **DNA KEY** to tailor Nutritional Intervention programs.
 3. Bi-directionally connect the **Patent Portal** and **Practitioner Portal** so practitioners can (a) submit invention disclosures and supporting evidence, and (b) track IP status directly in-workflow.
-

2) Explicitly Out of Scope (this SOW)

- AProof (proof-of-lawfulness modules, on-chain audit, ZK, consent proofs)
 - PIRTM (omics training pipelines, model registry/training)
 - OLHES (education/economic dashboards and learning loops)
 - HIPAA certification/audits and production EHR integrations (can be planned separately)
 - New ML model training; advanced analytics beyond deterministic & rules-based nutrition logic
-

3) Target Architecture (high-level)

3.1 Homirn Ring Integration

- **Mobile SDK (iOS/Android):** BLE pairing, offline queue, reconnect, packet de-dup/jitter handling.
- **Ingestion API + Device Registry:** Signed requests, rate/duplication controls, device → patient linkage.
- **Data Model:** Map telemetry to **FHIR R4** (Device, Observation, Patient, Provenance).
- **Observability:** Metrics on packet loss, latency, mapping errors.

3.2 DNA KEY Integration

- **Resolver Service:** Link DNA KEY → platform identities (patient/practitioner) and program entitlements.
- **Profile Enrichment:** Expose DNA-informed traits/flags that condition Nutritional Intervention rules.
- **Storage & Privacy:** Keys stored encrypted; no raw genetics in logs; rotatable key material.

3.3 Nutritional Intervention Service

- **Rules Engine (deterministic):** Thresholds and heuristics using ring telemetry + DNA KEY flags + basic questionnaire inputs.
- **Intervention Library:** Macronutrient targets, supplement guidance, hydration/sleep hygiene prompts; practitioner-editable.
- **Insights API:** Generates “why/what/next” messages and taskables for the Practitioner Portal.
- **Compliance Tracking:** Adherence markers, simple streaks, and follow-up reminders.

3.4 Patent Portal ↔ Practitioner Portal Wiring

- **SSO & Roles:** OIDC/SAML SSO; role mapping (Practitioner, Reviewer, IP Counsel).
 - **Connectors:** Patent Portal GraphQL/REST connector + webhook listener for status changes.
 - **Data Contracts:**
 - *From Practitioner → Patent:* Invention disclosure (title, abstract, claims draft), attachments (reports, anonymized telemetry summaries), inventors, witnesses, tags (DNA KEY, device, intervention).
 - *From Patent → Practitioner:* Status (draft, submitted, in review, filed), required actions, comments, docket dates.
 - **UI Surfaces in Practitioner Portal:**
 - **IP Hub widget:** Create/track disclosures, upload evidence, browse portfolio.
 - **Case Linking:** Associate disclosure to patient program or cohort (metadata only; no PHI exposure in Patent Portal).
 - **Evidence Packaging:** Automated generation of an **Invention Evidence Bundle** (anonymized ring features + intervention outcomes + timestamps) as a downloadable attachment within the Patent Portal record.
-

4) Deliverables

1. **Homir Ring SDK & Ingestion** (mobile SDKs, server ingestion, registry, FHIR mapping, metrics dashboard).
 2. **DNA KEY Resolver Service** (APIs, admin console, encryption at rest/in transit, rotation scripts).
 3. **Nutritional Intervention Service** (rules engine, intervention library CMS, Insights API, compliance tracking).
 4. **Patent Portal Connector** (API client, webhook listener, data contracts, error handling/retry).
 5. **Practitioner Portal Enhancements** (IP Hub widget, disclosure form, evidence bundler, status timeline).
 6. **Test Harness & Synthetic Data Suite** (SDK replay, mapping validator, rules tests, connector mocks).
 7. **Documentation & Handover** (runbooks, API specs, data dictionary, admin guides).
-

5) Acceptance Criteria (Phase 1)

- **SDK Reliability:** p95 reconnect $\leq 5s$; duplicate packet rate $< 0.5\%$; no data loss across app restarts/24h soak.
- **FHIR Validity:** $\geq 99.5\%$ of posted resources pass validation on synthetic sets.
- **DNA KEY Linking:** 99% successful resolution on eligible accounts; key rotation verified.
- **Nutrition Rules:** Deterministic rule pack executes $< 300ms$ p95; interventions render in Practitioner Portal within 2s end-to-end.

- **Patent Wiring:**
 - Create/edit/submit disclosure from Practitioner Portal; appears in Patent Portal within 5s.
 - Status updates/webhooks reflect in Practitioner Portal within 5s.
 - Evidence Bundle generated with correct metadata and attachments.
 - **Security & Privacy:** mTLS; least-privilege roles; zero secrets in repos; no PHI in logs.
-

6) Timeline & Milestones (indicative)

Week 1-2 — Discovery & Design

- Confirm ring BLE spec; draft data contracts; define DNA KEY fields; UX flows for IP Hub.

Week 3-6 — Build Track A (Device & Data)

- SDK + ingestion + FHIR mapping; metrics & alerts; basic Practitioner telemetry view.

Week 5-8 — Build Track B (DNA KEY & Nutrition)

- Resolver service + admin console; rules engine + intervention library; Insights API.

Week 7-10 — Build Track C (Patent/Practitioner Wiring)

- SSO & roles; connectors + webhooks; IP Hub widget; Evidence Bundle generator.

Week 11-12 — Hardening & UAT

- Synthetic end-to-end tests; performance tuning; documentation; handover.

Note: Parallel tracks reflect independent teams for SDK/Data, DNA/Nutrition, and Portals.

7) Dependencies & Assumptions

- Access to Homirrn Ring BLE specs/test firmware and a non-production firmware build.
 - Patent Portal provides stable API/webhooks and a sandbox tenant.
 - Practitioner Portal codebase access for widget integration.
 - DNA KEY schema/fields and initial mapping rules provided by CH Labs.
 - Synthetic datasets for validation; no PHI processed in this phase.
-

8) Security & Compliance (baseline)

- Encryption in transit (TLS 1.2+) and at rest; HSTS on web endpoints.
 - mTLS between ingestion and core data services; per-service secrets via a vault.
 - Centralized audit logging (no PHI); log retention policy documented.
 - RBAC with principle of least privilege; SSO for practitioner and patent users.
-

9) Handover Artifacts

- Architecture & sequence diagrams
- API specs (OpenAPI + examples)
- Rules pack source + tests
- Admin & runbooks
- Playbooks for support (triage, rollback, rotation)

10) Options (post-phase additions, not included in base)

- Advanced analytics/ML for nutrition outcomes
- Production EHR integrations and HIPAA audit readiness
- Re-introduction of PIRTM/AProof/OLHES when desired

11) Pricing (placeholder structure)

To be finalized with vendor after discovery.

Workstream	Fixed Fee	Notes
A. SDK & Ingestion (Ring → FHIR)	TBD	Mobile SDKs, ingestion, registry, mapping, metrics
B. DNA KEY Resolver & Admin	TBD	APIs, admin UI, encryption, rotation
C. Nutritional Intervention Service	TBD	Rules engine, library CMS, Insights API
D. Patent ↔ Practitioner Wiring	TBD	SSO, connectors, webhooks, IP Hub widget, Evidence Bundle
E. UAT, Hardening, Docs	TBD	Synthetic tests, perf tuning, handover
Total (Base Scope)	TBD	Excludes PIRTM/AProof/OLHES

12) RACI (snapshot)

- **Motivity (Vendor):** SDK & ingestion, DNA KEY resolver, Nutrition service, Patent/Practitioner connectors, tests, docs.
- **CH Labs (Client):** Architecture guardrails, DNA KEY schema/rules, patent sandbox access, practitioner UAT cohort.
- **Patent Portal Team (3rd-party):** API/webhook availability, SSO coordination, tenant setup.

Approval:

Name / Title / Date

Signature: _____